

Appendix A Extremal hypergraphs for $n = 5, 6, 7, 8$

For small cases, we obtained $\text{sat}_3(6, \text{Berge-}K_4) = 5$ and $\text{sat}_3(n, \text{Berge-}K_4) = n$ for $n = 5, 7, 8$ through computer calculations. All extremal hypergraphs are listed in Table 1. For hypergraphs with order n , we take the vertex set to be $\{0, 1, 2, \dots, n-1\}$. The last column of Table 1 denotes whether $\mathcal{T}$ can be added to each hypergraph: the pairs on which $\mathcal{T}$ can be added are provided where applicable, and “No” is recorded otherwise.

Table 1: Extremal hypergraphs by order

Order	No.	Hyperedges	Can add $\mathcal{T}$?
5	1	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (0, 3, 4)$	No
	2	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (1, 2, 4)$	No
	3	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (1, 2, 3)$	No
	4	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (2, 3, 4)$	$(0, 3), (1, 2)$
	5	$(0, 1, 2), (0, 1, 3), (0, 2, 3), (1, 2, 4), (1, 3, 4)$	$(2, 3)$
	6	$(0, 1, 2), (0, 1, 3), (0, 2, 4), (1, 3, 4), (2, 3, 4)$	$(0, 2), (0, 4)$
6	1	$(0, 1, 2), (0, 1, 3), (0, 2, 4), (1, 3, 4), (2, 3, 4)$	No
7	1	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 1, 5), (0, 2, 6), (2, 3, 4), (3, 5, 6)$	No
	2	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (0, 5, 6), (2, 3, 4), (2, 5, 6)$	$(0, 2)$
	3	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (0, 5, 6), (2, 3, 5), (2, 4, 6)$	$(0, 2)$
	4	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (1, 5, 6), (2, 3, 4), (2, 5, 6)$	$(1, 2)$
	5	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (1, 5, 6), (2, 3, 5), (2, 4, 6)$	$(1, 2)$
	6	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 2, 6), (1, 3, 5), (1, 4, 6)$	No
	7	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 2, 6), (1, 3, 5), (4, 5, 6)$	No
	8	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 2, 6), (1, 5, 6), (2, 3, 4)$	No
	9	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 2, 6), (1, 5, 6), (3, 4, 5)$	No
	10	$(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 3, 6), (1, 2, 6), (1, 4, 5)$	No

Order	No.	Hyperedges	Can add $\mathcal{T}$?
7	11	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 3, 6), (1, 4, 5), (2, 5, 6)	No
	12	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 3, 6), (2, 4, 5), (2, 4, 6)	No
	13	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 3, 6), (2, 4, 6), (3, 4, 5)	No
	14	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 5, 6), (1, 2, 6), (2, 3, 4)	No
	15	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 5, 6), (2, 3, 5), (2, 4, 6)	No
	16	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (1, 2, 6), (2, 3, 4), (2, 5, 6)	No
	17	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (1, 2, 6), (2, 3, 4), (3, 5, 6)	No
	18	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (1, 5, 6), (2, 3, 4), (2, 5, 6)	(1, 2)
	19	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 4), (2, 3, 6), (3, 5, 6)	No
	20	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 4), (2, 3, 6), (4, 5, 6)	No
	21	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 4), (3, 4, 6), (3, 5, 6)	No
	22	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 5), (2, 3, 6), (2, 4, 6)	No
	23	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 5), (2, 3, 6), (3, 4, 6)	No
	24	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 5), (2, 4, 6), (3, 5, 6)	No
	25	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 6), (3, 4, 5), (3, 4, 6)	No
	26	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (2, 3, 6), (3, 4, 5), (4, 5, 6)	No
	27	(0, 1, 2), (0, 1, 3), (0, 2, 3), (1, 2, 4), (1, 3, 4), (2, 5, 6), (3, 5, 6)	(2, 3)

Order	No.	Hyperedges	Can add $\mathcal{T}$?
7	28	(0, 1, 2), (0, 1, 3), (0, 2, 3), (1, 2, 4), (1, 3, 5), (2, 5, 6), (3, 4, 6)	(2, 3)
	29	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 4), (0, 5, 6), (1, 2, 5), (1, 3, 6)	No
	30	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 4), (0, 5, 6), (1, 2, 5), (3, 4, 6)	No
	31	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 4, 6), (1, 3, 4), (1, 5, 6)	No
	32	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 4, 6), (1, 3, 4), (4, 5, 6)	No
	33	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 4, 6), (1, 3, 6), (4, 5, 6)	No
	34	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 4, 6), (1, 4, 6), (1, 5, 6)	No
	35	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 4, 6), (1, 5, 6), (3, 4, 5)	No
	36	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (1, 2, 5), (1, 3, 6), (4, 5, 6)	No
	37	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (1, 2, 6), (1, 4, 5), (4, 5, 6)	No
	38	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (2, 3, 4), (2, 5, 6), (3, 4, 6)	(2, 3)
	39	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 2, 5), (1, 4, 6), (3, 5, 6)	No
	40	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 2, 5), (3, 4, 5), (3, 4, 6)	No
	41	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (1, 5, 6), (2, 3, 4)	(0, 1)
	42	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (2, 3, 4), (3, 5, 6)	(0, 3)
	43	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (2, 3, 5), (2, 3, 6)	(0, 3)
	44	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (2, 3, 5), (3, 5, 6)	(0, 3)

Order	No.	Hyperedges	Can add $\mathcal{T}$?
7	45	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (2, 4, 5), (3, 4, 6)	(0, 4)
	46	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (1, 3, 4), (2, 4, 5), (4, 5, 6)	(0, 4)
	47	(0, 1, 2), (0, 1, 3), (0, 2, 4), (1, 3, 4), (2, 3, 5), (2, 3, 6), (4, 5, 6)	No
	48	(0, 1, 2), (0, 1, 3), (0, 2, 4), (1, 3, 5), (2, 3, 6), (2, 5, 6), (3, 4, 6)	No
8	1	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (0, 2, 5), (0, 6, 7), (3, 4, 5), (3, 6, 7)	(0, 3)
	2	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 3), (1, 2, 5), (1, 6, 7), (3, 4, 5), (3, 6, 7)	(1, 3)
	3	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 6, 7), (1, 3, 5), (2, 4, 5), (5, 6, 7)	(0, 5)
	4	(0, 1, 2), (0, 1, 3), (0, 1, 4), (0, 2, 5), (0, 6, 7), (2, 3, 6), (2, 3, 7), (3, 4, 5)	No
	5	(0, 1, 2), (0, 1, 3), (0, 2, 3), (0, 4, 5), (0, 6, 7), (1, 2, 4), (1, 3, 5), (1, 6, 7)	(0, 1)
	6	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 6, 7), (1, 2, 5), (1, 4, 6), (1, 4, 7)	No
	7	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 6, 7), (1, 2, 5), (3, 4, 5), (5, 6, 7)	(0, 5)
	8	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 6, 7), (1, 2, 6), (1, 3, 7), (1, 4, 5)	No
	9	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 6, 7), (1, 2, 6), (1, 4, 5), (1, 4, 7)	No
	10	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 3, 5), (0, 6, 7), (1, 2, 6), (1, 4, 7), (5, 6, 7)	No
	11	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (0, 5, 7), (1, 3, 4), (2, 3, 6), (3, 5, 7)	(0, 3)
	12	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (0, 5, 7), (1, 3, 4), (2, 4, 5), (4, 6, 7)	(0, 4)
	13	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (0, 5, 7), (1, 3, 4), (2, 4, 6), (4, 5, 7)	(0, 4)

Order	No.	Hyperedges	Can add $\mathcal{T}$?
8	14	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (0, 5, 7), (1, 3, 5), (2, 3, 6), (3, 4, 7)	(0, 3)
	15	(0, 1, 2), (0, 1, 3), (0, 2, 4), (0, 5, 6), (0, 5, 7), (1, 3, 6), (2, 3, 7), (3, 4, 5)	(0, 3)
	16	(0, 1, 2), (0, 1, 3), (0, 4, 5), (0, 4, 6), (1, 4, 7), (1, 5, 6), (2, 4, 7), (3, 5, 7)	No
	17	(0, 1, 2), (0, 1, 3), (0, 4, 5), (0, 4, 6), (1, 5, 7), (2, 3, 5), (2, 4, 7), (3, 5, 6)	No